

Detailed archival supplement: A self-contained proof of primitive trace-one quadratics over $\mathbb{F}_{2^k}$

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Abstract

This supplement is an archival, expanded version of the even-characteristic PTE proof used in Paper A. It contains the complete paper proof, additional derivations of the Gauss-sum and Ramanujan-sum ingredients, the full finite certificate for $1 \leq k \leq 29$, the exact factorizations of $2^{2^k} - 1$, and reproducibility and formalization notes. The theorem is known: the statement is the $n = 2, t = 1$ case of Cohen (1990, Theorem 1, p. 1), while Moreno (1989) proved the trace-one quadratic case earlier. Cohen (2003, Theorem 1.5, p. 170) gives an equivalent square-order formulation. No priority is claimed.

1 Status and scope

The purpose of this note is to keep a detailed proof and computation archive for the following statement. It is not a claim of new priority for the theorem itself. Paper A uses a compressed version of this material in its Appendix B.

2 The theorem and notation

Theorem 1. *For every $k \geq 1$, there exists*

$$\beta \in \mathbb{F}_{2^{2k}}^*, \quad \text{ord}(\beta) = 2^{2^k} - 1, \quad \text{Tr}_{\mathbb{F}_{2^{2k}}/\mathbb{F}_{2^k}}(\beta) = 1.$$

Equivalently, there is an irreducible quadratic $X^2 + X + a \in \mathbb{F}_{2^k}[X]$ whose roots are primitive.

3 Detailed proof

Proof of Theorem 1. The statement is classical (see [3] and [2, Theorem 1, p. 1]); it also follows from the equivalent square-order formulation in [1, Theorem 1.5, p. 170]); we give a self-contained proof in the even-characteristic case used above. This is a reconstruction of known results (Moreno; Cohen), and no priority is claimed; it is included only to make the paper self-contained. Put $q = 2^k$, $L = \mathbb{F}_{q^2}$, $K_0 = \mathbb{F}_q$, and $M = q^2 - 1 = (q - 1)(q + 1)$. Let

$$E_1 = \{x \in L : x^q + x = 1\}.$$

The trace map $L \rightarrow K_0$ is surjective and K_0 -linear, so $|E_1| = q$. If $x \in E_1$, then $x \notin K_0$ and its minimal polynomial over K_0 is $X^2 + X + xx^q$. Hence it is enough to prove that E_1 contains an element of order M .

Let ψ be the nontrivial additive character of $\mathbb{F}_2$, and put $\psi_L(x) = \psi(\text{Tr}_{L/\mathbb{F}_2}(x))$ and $\psi_{K_0}(a) = \psi(\text{Tr}_{K_0/\mathbb{F}_2}(a))$. For a multiplicative character χ of L^* define

$$g_L(\chi) = \sum_{x \in L^*} \chi(x) \psi_L(x), \quad g_{K_0}(\chi) = \sum_{a \in K_0^*} \chi(a) \psi_{K_0}(a), \quad S(\chi) = \sum_{x \in E_1} \chi(x).$$

If χ is nontrivial on L^* , then $|g_L(\chi)| = q$; if $\chi|_{K_0^*}$ is nontrivial, then $|g_{K_0}(\chi)| = \sqrt{q}$. Indeed, the usual squared-modulus calculation gives

$$|g_L(\chi)|^2 = \sum_{t \in L^*} \chi(t)^{-1} \sum_{a \in L^*} \psi_L(a(1-t)) = q^2,$$

and the same argument in K_0 gives q .

We first compute $S(\chi)$. For $\alpha \in K_0$ set $E_\alpha = \{x \in L : x^q + x = \alpha\}$; these sets partition L , and $E_\alpha = \alpha E_1$ for $\alpha \neq 0$. If $\chi^{q+1} \neq 1$, then the restriction of χ to K_0^* is nontrivial; hence

$$g_L(\chi) = \sum_{\alpha \in K_0} \psi_{K_0}(\alpha) \sum_{x \in E_\alpha} \chi(x) = S(\chi) g_{K_0}(\chi),$$

so $|S(\chi)| = q/\sqrt{q} = \sqrt{q}$.

Suppose now that $\chi^{q+1} = 1$ and $\chi \neq 1$. Then $\chi|_{K_0^*} = 1$. Let $H = \{h \in L^* : h^{q+1} = 1\}$; since $\gcd(q-1, q+1) = 1$, the group L^* is the direct product $K_0^* \times H$. Write $x = ch$ with $c \in K_0^*$ and $h \in H$. The condition $x^q + x = 1$ becomes $c(h + h^{-1}) = 1$; thus $h \neq 1$ and $c = (h + h^{-1})^{-1}$. This gives a bijection between $H \setminus \{1\}$ and E_1 , and therefore

$$S(\chi) = \sum_{h \in H, h \neq 1} \chi(h) = \sum_{h \in H} \chi(h) - 1 = -1,$$

because $\chi|_H$ is nontrivial.

We shall use the primitive-element indicator

$$\mathbf{1}_{\text{ord}(x)=M} = \frac{\varphi(M)}{M} \sum_{\delta|M} \frac{\mu(\delta)}{\varphi(\delta)} \sum_{\text{ord}(\chi)=\delta} \chi(x). \quad (1)$$

To see this, fix a generator g of L^* and write $x = g^a$. The left side is the indicator of $\gcd(a, M) = 1$. Its Fourier coefficient at a character of order δ is the Ramanujan sum $c_M(j) = \mu(\delta)\varphi(M)/\varphi(\delta)$; Fourier inversion gives (1). (The displayed identity for $c_M(j)$ follows by Möbius inversion.)

Assume for contradiction that E_1 has no element of order M . Summing (1) over E_1 gives

$$0 = \frac{\varphi(M)}{M} \sum_{\delta|M} \frac{\mu(\delta)}{\varphi(\delta)} S_\delta, \quad S_\delta = \sum_{\text{ord}(\chi)=\delta} S(\chi).$$

Let $s = \omega(M)$ and $s' = \omega(q+1)$ be the numbers of distinct prime divisors. The $\delta = 1$ term is q . If $\delta \mid q+1$ and $\delta > 1$, then each character of order δ satisfies $\chi^{q+1} = 1$, so the corresponding contribution is $-\mu(\delta)$; summing over $\delta \mid q+1$, $\delta > 1$, gives 1. For the remaining squarefree divisors $\delta \nmid q+1$, the character sums are bounded by $\sqrt{q}$, and their number is at most $2^s - 2^{s'}$. Consequently,

$$q+1 \leq (2^s - 2^{s'})\sqrt{q}. \quad (\text{A.2})$$

Since $q+1 > \sqrt{q}$, this implies

$$2^s > 2^{k/2}, \quad \text{so} \quad s > k/2.$$

In particular $k < 2s$. Let $p_1 < p_2 < \dots$ be the primes and $P_s = p_1 \cdots p_s$. Since $M = 2^{2k} - 1$ has exactly s distinct prime divisors, $M \geq P_s$. We have

$$P_{16} = 32589158477190044730 > 2^{64}.$$

For $n \geq 16$, the ratio

$$\frac{P_{n+1}/2^{4(n+1)}}{P_n/2^{4n}} = \frac{p_{n+1}}{16}$$

is greater than 1, because $p_{n+1} \geq 59$. Hence $P_s > 2^{4s}$ for every $s \geq 16$. But $k < 2s$ gives

$$M = 2^{2k} - 1 < 2^{2k} < 2^{4s},$$

so $s \geq 16$ is impossible. Thus $s \leq 15$ and $k < 2s \leq 30$, i.e. $k \leq 29$.

It remains to cover this finite range. In the table below, m_k is an irreducible polynomial of degree k over $\mathbb{F}_2$, encoded by the binary coefficient vector of the displayed hexadecimal integer; thus $K_0 = \mathbb{F}_2[x]/(m_k)$. The element $a_k \in K_0$ is encoded similarly. For each row let t be a root of $X^2 + X + a_k$ in $K_0[t]/(X^2 + X + a_k)$. The deterministic verifier checks that m_k is irreducible, that $\text{Tr}_{K_0/\mathbb{F}_2}(a_k) = 1$, that $t^{2^k} = t + 1$, and that $t^M = 1$ while $t^{M/p} \neq 1$ for every prime $p \mid M$. The last two checks show that $\text{ord}(t) = M$; the third shows that $\text{Tr}_{L/K_0}(t) = 1$. A verifier for the table, `verify_k1_29.py`, is included as an ancillary file; its SHA-256 hash is 042CE7F74EDFF13ADE5C6285E42CB3D15491FC25C65B99BC670D256047EE8D9F.

k	m_k	a_k
1	0x3	0x1
2	0x7	0x3
3	0xb	0x5
4	0x13	0xd
5	0x25	0x13
6	0x43	0x34
7	0x83	0x47
8	0x11b	0xc8
9	0x203	0x2b
10	0x409	0xcb
11	0x805	0x525
12	0x1009	0x39e
13	0x201b	0x1ec4
14	0x4021	0x3717
15	0x8003	0x62bf
16	0x1002b	0x1e6e
17	0x20009	0x1155b
18	0x40009	0x2ee84
19	0x80027	0xb9af
20	0x100009	0x6aee2
21	0x200005	0x15750b
22	0x400003	0x24a150
23	0x800021	0x6c92f9
24	0x100001b	0x389ec4
25	0x2000009	0x6801ad
26	0x400001b	0x2707d2e
27	0x8000027	0x6e9ab1e
28	0x10000003	0xfac7cd2
29	0x20000005	0xc69b55e

The table therefore supplies the required element for every $k \leq 29$. Combining this with the contradiction for $k \geq 30$ proves the theorem. $\square$

4 Supplementary derivations

4.1 Gauss-sum squared modulus

For a nontrivial multiplicative character χ of L^* , put

$$g_L(\chi) = \sum_{x \in L^*} \chi(x) \psi_L(x).$$

Then

$$|g_L(\chi)|^2 = \sum_{t \in L^*} \chi(t)^{-1} \sum_{a \in L^*} \psi_L(a(1-t)).$$

For $t = 1$, the inner sum is $|L| - 1$; for $t \neq 1$, it is -1 . Since $\sum_{t \in L^*} \chi(t)^{-1} = 0$, the double sum is $|L|$. This is the calculation used in the proof.

4.2 Ramanujan-sum identity

Let M be the order of L^* , fix a generator g , and write $x = g^a$. For a character $\chi_j(g) = e^{2\pi i j/M}$, the Fourier coefficient of the indicator of $\gcd(a, M) = 1$ is

$$c_M(j) = \sum_{\substack{a \bmod M \\ \gcd(a, M) = 1}} e^{-2\pi i j a/M}.$$

With $\delta = M/\gcd(M, j)$, the identity

$$c_M(j) = \sum_{d|\gcd(M, j)} d \mu(M/d) = \mu(\delta) \frac{\varphi(M)}{\varphi(\delta)}$$

follows by Möbius inversion. Fourier inversion then gives the primitive-element indicator used in the paper. This is the detailed form of the formula cited in the main proof.

4.3 Why the finite checks suffice

For each row of the certificate, let t be a root of $X^2 + X + a_k$ in $K_0[t]/(X^2 + X + a_k)$. The verifier checks:

1. m_k is irreducible of degree k , so $K_0 = \mathbb{F}_2[x]/(m_k) \cong \mathbb{F}_{2^k}$;
2. $\text{Tr}_{K_0/\mathbb{F}_2}(a_k) = 1$, so $X^2 + X + a_k$ is irreducible over K_0 ;
3. $t^{2^k} = t + 1$, hence $\text{Tr}_{L/K_0}(t) = 1$;
4. $t^M = 1$ and $t^{M/p} \neq 1$ for every prime $p \mid M$.

The last two conditions imply $\text{ord}(t) = M = 2^{2k} - 1$. Thus t is the required primitive trace-one element.

5 Full finite certificate

The following table gives the hexadecimal polynomial encodings used by the verifier:

$$m_k = \sum_{i=0}^k m_{k,i} x^i, \quad a_k = \sum_{i=0}^{k-1} a_{k,i} x^i,$$

with the hexadecimal integer encoding the binary coefficient vector.

k	m_k	a_k	$\omega(2^{2k} - 1)$
1	0x3	0x1	1
2	0x7	0x3	2
3	0xB	0x5	2
4	0x13	0xD	3
5	0x25	0x13	3
6	0x43	0x34	4
7	0x83	0x47	3
8	0x11B	0xC8	4
9	0x203	0x2B	4
10	0x409	0xCB	5
11	0x805	0x525	4
12	0x1009	0x39E	6
13	0x201B	0x1EC4	3
14	0x4021	0x3717	6
15	0x8003	0x62BF	6
16	0x1002B	0x1E6E	5
17	0x20009	0x1155B	3
18	0x40009	0x2EE84	8
19	0x80027	0xB9AF	3
20	0x100009	0x6AEE2	7
21	0x200005	0x15750B	6
22	0x400003	0x24A150	7
23	0x800021	0x6C92F9	4
24	0x100001B	0x389EC4	9
25	0x2000009	0x6801AD	7
26	0x400001B	0x2707D2E	7
27	0x8000027	0x6E9AB1E	6
28	0x10000003	0xFAC7CD2	8
29	0x20000005	0xC69B55E	6

The exact prime-power factorizations used by the verifier are:

k	$\omega(M)$	factorization of $M = 2^{2k} - 1$
1	1	3
2	2	$3 \cdot 5$
3	2	$3^2 \cdot 7$

k	$\omega(M)$	factorization of $M = 2^{2^k} - 1$
4	3	$3 \cdot 5 \cdot 17$
5	3	$3 \cdot 11 \cdot 31$
6	4	$3^2 \cdot 5 \cdot 7 \cdot 13$
7	3	$3 \cdot 43 \cdot 127$
8	4	$3 \cdot 5 \cdot 17 \cdot 257$
9	4	$3^3 \cdot 7 \cdot 19 \cdot 73$
10	5	$3 \cdot 5^2 \cdot 11 \cdot 31 \cdot 41$
11	4	$3 \cdot 23 \cdot 89 \cdot 683$
12	6	$3^2 \cdot 5 \cdot 7 \cdot 13 \cdot 17 \cdot 241$
13	3	$3 \cdot 2731 \cdot 8191$
14	6	$3 \cdot 5 \cdot 29 \cdot 43 \cdot 113 \cdot 127$
15	6	$3^2 \cdot 7 \cdot 11 \cdot 31 \cdot 151 \cdot 331$
16	5	$3 \cdot 5 \cdot 17 \cdot 257 \cdot 65537$
17	3	$3 \cdot 43691 \cdot 131071$
18	8	$3^3 \cdot 5 \cdot 7 \cdot 13 \cdot 19 \cdot 37 \cdot 73 \cdot 109$
19	3	$3 \cdot 174763 \cdot 524287$
20	7	$3 \cdot 5^2 \cdot 11 \cdot 17 \cdot 31 \cdot 41 \cdot 61681$
21	6	$3^2 \cdot 7^2 \cdot 43 \cdot 127 \cdot 337 \cdot 5419$
22	7	$3 \cdot 5 \cdot 23 \cdot 89 \cdot 397 \cdot 683 \cdot 2113$
23	4	$3 \cdot 47 \cdot 178481 \cdot 2796203$
24	9	$3^2 \cdot 5 \cdot 7 \cdot 13 \cdot 17 \cdot 97 \cdot 241 \cdot 257 \cdot 673$
25	7	$3 \cdot 11 \cdot 31 \cdot 251 \cdot 601 \cdot 1801 \cdot 4051$
26	7	$3 \cdot 5 \cdot 53 \cdot 157 \cdot 1613 \cdot 2731 \cdot 8191$
27	6	$3^4 \cdot 7 \cdot 19 \cdot 73 \cdot 87211 \cdot 262657$
28	8	$3 \cdot 5 \cdot 17 \cdot 29 \cdot 43 \cdot 113 \cdot 127 \cdot 15790321$
29	6	$3 \cdot 59 \cdot 233 \cdot 1103 \cdot 2089 \cdot 3033169$

6 Reproducibility

The deterministic verifier is `verify_k1_29.py`, stored in the Paper D `verification/` directory and also included in the submission package. It uses only integer and polynomial arithmetic and requires neither network access nor SymPy. Run:

```
python verification/verify_k1_29.py
```

The expected final line is:

```
All certificates verified.
```

The SHA-256 of the verifier is

```
042CE7F74EDFF13ADE5C6285E42CB3D15491FC25C65B99BC670D256047EE8D9F
```

The verifier was also independently reimplemented during the red-team audit; all $k = 1, \dots, 29$ rows passed irreducibility, trace, Frobenius, and exact-order checks.

7 Formalization boundary

The affine group-generation argument and selected algebraic lemmas are formalized in Lean 4 with mathlib. A conditional Lean file formalizes the field-level Frobenius identities and the order-theoretic τ -collapse once an explicit order interface is supplied. The analytic PTE proof and the finite $k \leq 29$ certificate in this note are not Lean-formalized. This is a disclosed boundary, not a hidden dependency.

8 Historical attribution

The theorem is classical. The exact historical theorem used here is the $n = 2, t = 1$ case of [2, Theorem 1, p. 1]. Moreno [3] proved the trace-one quadratic case earlier. Cohen [1, Theorem 1.5, p. 170] gives an equivalent square-order formulation. These references are historical attributions only; the proof above is self-contained.

References

- [1] S. D. Cohen, *The orders of related elements of a finite field*, Ramanujan J. 7 (2003) 169–183, doi:10.1023/A:1026295128600.
- [2] S. D. Cohen, *Primitive elements and polynomials with arbitrary trace*, Discrete Math. 83 (1990) 1–7.
- [3] O. Moreno, *On the existence of a primitive quadratic of trace 1 over $GF(p^m)$* , J. Combin. Theory Ser. A 51 (1989) 104–110.